ResearchDAO Whitepaper

Decentralized research funding and governance. Version 1.0. Unaudited reference design.

1. Overview

ResearchDAO is an onchain platform where researchers publish proposals, communities and reviewers decide on them, funders contribute BOT or USDT, and money is released milestone by milestone. All decisions and payments leave a public record; confidential research does not.

2. Governance

Each funding round is configured by DAO administrators: payment token, per-proposal budget cap, voting period, funding period, wallet-vote quorum and the number of reviewer approvals required. A proposal is funded only if (a) the voting period has ended, (b) quorum is met, (c) yes votes exceed no votes, and (d) enough authorized reviewers approved it. Any reviewer rejection ends the proposal immediately. Anyone can finalize an expired vote, so no party can stall a decision.

Wallet voting is not one-person-one-vote. The contract counts one vote per wallet address per proposal and prevents the same address voting twice. It cannot tell whether many wallets belong to one person. Mitigations are the quorum, mandatory reviewer approval, and optional future identity attestations (for example verified credentials). Community votes should be read as a wallet-level signal of support, not a measure of verified human opinion.

3. Funding and treasury

Contributions are accepted only while a proposal is in its funding phase and never beyond the proposal budget. Funds are held by the treasury contract and tracked per proposal and per funder. When the budget is fully raised the proposal becomes Active. If the funding window expires first, the proposal is marked Failed and every funder may reclaim their contribution.

4. Milestones and disbursement

Budgets are split into 1 to 20 milestones whose amounts must sum to the budget. The researcher submits completion evidence (a hash and a public URI) in order; authorized reviewers verify; once the round's approval threshold is reached an administrator authorizes the payment. A milestone can be paid exactly once, total payments can never exceed funds raised, and a proposal completes when all milestones are paid. Every payment emits a public Disbursed event.

5. Cancellation and refunds

Researchers or administrators may cancel a proposal that is not yet final. Rejected, Failed and Cancelled proposals allow each funder to claim a pro-rata share of undisbursed funds: contribution x (raised - paid) / raised. Each funder can claim once.

6. Transparency

Onchain records include researcher registrations, proposals, votes, reviews, contributions, evidence hashes, verifications, disbursements and refunds. Anyone can audit the full lifecycle from events and

view functions.

7. Privacy

Public blockchains are permanent and readable by everyone. ResearchDAO therefore stores only titles, categories, public summaries, URIs, milestone descriptions and cryptographic hashes. Confidential research, unpublished sensitive data, personal information and restricted datasets must never be written onchain or linked from public URIs without access control. To prove integrity of private material, publish only its hash and share the material off-chain under appropriate agreements. Researcher profiles should contain professional, public information only.

8. Security

Risk	Control
Double voting	Per-wallet, per-proposal flag
Unauthorized funding	Funding allowed only in Funding status, before deadline
Over-budget payment	Contributions capped at budget; payments capped at raised funds
Duplicate milestone payment	Milestone state machine; Paid is terminal
Reentrancy	nonReentrant guards, checks-effects-interactions, SafeERC20
Admin compromise	Use a multisig/timelock as admin; reviewers are separate roles
Sybil voting	Quorum + reviewer approval; identity attestations recommended
Token risk	Whitelist audited tokens per round; mind decimals (USDT is often 6)

The reference contracts are unaudited. Obtain an independent audit and run a testnet pilot before holding real funds. Administrators hold significant power (reviewer set, payment authorization); decentralizing them is a roadmap item.

9. Limitations and roadmap

Single-contract design for clarity; future versions may split modules, add identity-gated or token-weighted voting, reviewer rotation, dispute resolution and multi-token rounds.